The Norns III: Seeding Simulated-Annealing Placement with Provenance and Recovered Clusters How hot can the anneal start before it destroys a good initial placement?

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***Abstract*—Simulated annealing is the workhorse of FPGA placement, and it starts from an initial placement that the anneal then destroys and rebuilds. What if the initial placement carried real information – the HDL hierarchy a provenance-tracked front end kept, or the clusters a graph analysis recovered from the flat netlist? This paper asks whether such seeds improve placement, and, because the anneal's early phase can erase any seed, at what initial temperature the seed's advantage dies. The seeds come from Odin III's provenance hierarchy and from Loki III's L2 clusterings (Louvain, spectral, affinity propagation, k-means); the die-level arrangement of clusters comes from a cut-weighted quotient-graph layout; and the seeds enter VPR through its initial-placement and floorplanning-constraint interfaces, with one additive, default-off patch for checkpointing, following RLPlace. We report seed retention against the annealing temperature, quality of results against an unseeded baseline, and the crossover temperature at which each seed stops helping – the recipe a clustering-based initial placement should be used with.**

***Index Terms*—FPGA placement, simulated annealing, provenance, clustering, VTR**

I. Introduction

VPR's simulated-annealing placer starts from an initial placement and anneals it: the schedule begins hot enough to move anything, and the early phase rearranges the circuit wholesale. The question this paper asks is whether the initial placement can be made to *mean* something, and how much of that meaning survives the anneal. Gharibian et al. showed that system-level structure recovered from a flat netlist correlates with placement locality [**?**], [**?**], [**?**]. Odin III keeps the provenance link that makes recovery unnecessary for the hierarchy [**?**], and Loki III recovers clusters from the flat netlist without a language model. RLPlace [**?**] showed that the anneal itself can be improved – directed moves chosen by a reinforcement-learning agent and checkpointing that prevents oscillation – starting from VTR's analytic placement. This paper connects those lines: the seed carries the structure, the anneal is measured for how much of it the seed can keep, and the temperature at which the advantage dies is reported as data, not tuned away.

The contributions are: (1) a seeding methodology – provenance hierarchy and four recovered clusterings, arranged on the die by a quotient-graph layout, entering VPR through its initial-placement and constraint interfaces with a default-off checkpointing patch; (2) the crossover sweep, which calibrates the highest useful initial temperature for each seed against the best placement any seed achieves; (3) the retention and quality numbers per seed, with the full-seed versus inter-cluster-only split that separates the value of the coarse layout from the fine one.

II. Background

A. Simulated annealing placement and RLPlace

VPR places by simulated annealing from an analytic initial placement [**?**], [**?**]. The anneal's moves are mostly random; its early phase can undo any structure the start carried. RLPlace improves the anneal rather than the start: a K-armed bandit selects among directed move types (simple moves and swaps, bounded centroid, median, critical-path moves), and checkpointing periodically saves the best placement so oscillation cannot throw it away [**?**]. Our experiment keeps the anneal as-is except for one default-off checkpointing flag, and changes the start instead; RLPlace's checkpointing prediction – that save-best/restore extends the useful temperature range – is one of the hypotheses we test.

B. The seeds

The provenance hierarchy (Urd) is the module structure Odin III records on every netlist object; it needs no recovery. The recovered clusters (Verdandi) are Loki III L2 partitions: Louvain, spectral, affinity propagation and k-means at several k, scored in the Loki paper against the known hierarchy. The two seed families differ exactly in the question's terms: provenance is the structure the designer wrote, clusters are the structure the flat netlist still shows.

III. Methodology

A. Seed construction

Each seed is a partition of the placed blocks. Its die-level arrangement comes from the quotient graph: contract each cluster to a node, weight edges by inter-cluster net cuts, lay

TABLE I
GEOMEAN QUALITY RATIOS VS THE UNSEEDED BASELINE, PER SEED AND INJECTION (SOFT INITIAL PLACEMENT, HARD CONSTRAINTS).

Seed	Wirelength	Delay	Chan. width	Runtime
Urd (hierarchy)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**
Louvain	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**
Spectral	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**
Affinity propagation	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**
k-means (k, $2k$, $4k$)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**
Inter-cluster only	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**

the quotient out by a coarse force-directed solve, and scale it to the die. Inside a cluster, blocks take the seed's internal order – or a shuffle, in the inter-cluster-only arm, so the two arms separate the coarse layout's value from the fine one.

B. Injection

Seeds enter VPR two ways, both already in the pinned build (v8.0.0+18416): soft, as an initial placement file (`--read_initial_place_file`) that the anneal is free to move away from; and hard, as floorplanning constraints (`--read_vpr_constraints`) with one PartitionRegion per cluster, which the anneal cannot violate at any temperature. `--write_initial_place_file` dumps VPR's own analytic start for the baseline study.

C. The crossover sweep

The calibration loop follows the data, in order. First, run every seed and keep the best final placement P^* (and which seed made it). Second, cold-start a fresh anneal from P^* and sweep the initial temperature upward; the temperature at which quality falls to the unseeded baseline is the crossover – the highest useful start temperature for a cluster-derived placement. Third, run every seed at a temperature just above its crossover against an unseeded run at the same temperature: whether the seed still wins there says whether its information is coarse (which anneals away last) or fine. The checkpointing arm repeats the sweeps with save-best/restore enabled; per RLPlace, the crossover should move hotter.

D. Metrics and data

Per run: seed retention (fraction of blocks still inside their seed cluster's region at exit; mean displacement from the seed coordinate), routed wirelength, critical-path delay, minimum channel width, placement runtime, and the per-temperature cost curve. Designs are the 15 VTR benchmarks that fit the development machine, smallest first, one run at a time under the heavy lock. Every number in this paper traces to a kept record; large data keeps its scripts.

IV. RESULTS

The numbers land as the runs complete; the retention-versus-temperature curves are the experiment's central figure and will be added here.

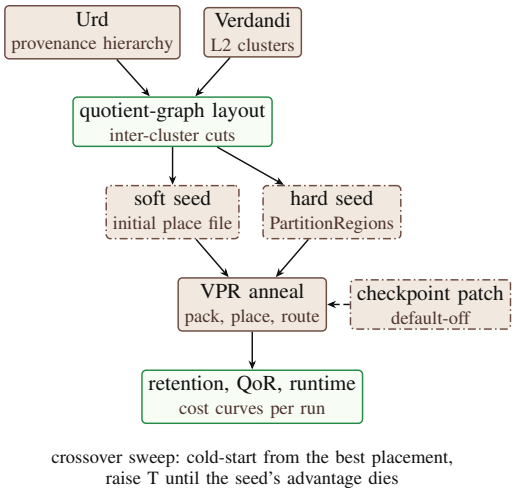

Fig. 1. The Norns III experiment. Seeds from provenance and recovered clusters, arranged by a quotient-graph layout, enter VPR softly (initial placement) or hard (constraints). The crossover sweep cold-starts from the best placement and raises the initial temperature until the seed's advantage dies; the checkpointing arm (one default-off patch) tests whether save-best/restore extends the range.

TABLE II
THE CROSSOVER SWEEP: HIGHEST INITIAL TEMPERATURE AT WHICH THE COLD-START FROM THE BEST PLACEMENT STILL MATCHES ITS QUALITY, AND RETENTION AT THAT TEMPERATURE, PER SEED.

Seed	Crossover T	Retention
Urd (hierarchy)	**[TODO:]**	**[TODO:]**
Louvain	**[TODO:]**	**[TODO:]**
Spectral	**[TODO:]**	**[TODO:]**
Affinity propagation	**[TODO:]**	**[TODO:]**
k-means	**[TODO:]**	**[TODO:]**

V. DISCUSSION

[TODO: Three observations once the runs land: which seed family wins and whether it is coarse or fine information; whether the checkpointing arm moves the crossover hotter, as RLPlace predicts; and what the safe temperature recipe is for a cluster-derived start.]

VI. CONCLUSION

[TODO: Written from the results.]